CHIMERA-QRC: A Pre-Registered, Adversarially-Controlled Study of Quantum Reservoir Computing for S&P 500 Realized-Volatility Forecasting

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Abstract. We contribute a **pre-registered, adversarially-controlled evaluation instrument** for quantum reservoir computing — reproducible **offline in one command** (`cli.py verify`), with every hardware prediction bound to a prior commit by a **hash-preimage commitment** — and apply it to S&P 500 realized-volatility forecasting across **four devices from three QPU vendors** (IonQ, IQM, Rigetti). Two measurements stand on their own: an empirical **quantum-complexity metric** (the reservoir's MPS bond dimension is essentially full, **χ_eff ≈ 2^(n/2)**, marking the classical-simulability boundary) and a cross-platform **coherence-budget wall** (Fig. 3) whose readout-corrected fingerprint separates **coherent circuit routing from depolarizing/damping noise**, each regime verdict **9.7–23.9σ** beyond shot noise. Applied honestly, the instrument returns a **well-characterized negative**: at simulable scale (≤16 qubits) the reservoir is *distinct but not more accurate* than the strongest fair baselines — **HAR-X**, size-matched ESN/RFF, under HAC-DM + Holm + MCS (**H0 refuted**) — robust across a second **Heisenberg** reservoir family (not Ising-specific) and a second (weather) domain, and our program **falsified four of its own pre-registered predictions** under controls (a same-session cross-seed control shows the n=8 scrambling is *seed-0-instance-specific, not a size law*). **Desk takeaway:** vol-timing a simple RV forecast ≈halves the 2008 drawdown; quantum adds no economic value at this scale.

1. Focus, track selection, and problem framing

We target **Track A: financial volatility** — one-step-ahead forecasting of S&P 500 **realized variance (RV)** and the tracking of volatility **regime transitions**, the domain where JonesTrading's risk, allocation and derivatives pricing live. RV is long-memory, multi-scale, regime-switching and non-Gaussian — the structure reservoir computing exploits. Our central yardsticks are the Echo State Network (ESN, the classical analog of QRC) and the strongest econometric models (HAR — *Heterogeneous AutoRegressive*, Corsi 2009 — and, critically for Phase 3, **HAR-X**: HAR augmented with the same realized-measure features we feed the quantum reservoir). We also report GARCH/GJR-GARCH, AR(3), persistence, LSTM, and an RFF/RBF kernel, and we implement the challenge's common **MNIST** benchmark on the identical engine.

This paper's honest thesis. We pre-registered H0 and *attacked our own result* with fair controls: the dominant accuracy lever is *which features are encoded*, not quantum nonlinearity. What survives for the reservoir — kernel distinctness, lower seed variance, full-rank entanglement (classical-simulation hardness) — is *necessary, not sufficient* for advantage. **Vs. prior work.** The closest study (Li et al. 2025/26 — QRC for realized volatility) presents the approach as a competitive, noise-resilient *proof-of-concept*; we add the control-hardened, pre-registered evaluation it omits — the decisive **HAR-X** control, HAC-DM/Holm/MCS, and explicit capability and per-layer-noise audits — turning an encouraging proof-of-concept into a falsifiable result.

2. QRC architecture (Fig. 1)

CHIMERA is a delay-embedding quantum reservoir. Inputs are angle-encoded `RY(π·x)`, one value per qubit, onto `|0…0⟩`; the state evolves under a fixed transverse-field **Ising** Hamiltonian `H = Σ_{i<j} J_ij Z_iZ_j + h_x Σ_i X_i` (`h_x=1`; `J` a **random ≈50%-connected** graph, `connectivity=0.5`; `U=exp(-iHτ)`, `τ=2`); single- and pairwise Pauli-Z expectations `⟨Z_i⟩,⟨Z_iZ_j⟩` form the feature vector. A **ridge head consumes these features concatenated with a linear block of the same inputs (incl. HAR)**, so any reservoir gain is genuine nonlinearity *beyond* the linear span of identical information. The identical inputs feed the classical controls, isolating quantum-vs-classical. Four mechanisms extend the core (multi-scale τ-bank; RZ measurement feedback; regime-adaptive Ising↔Heisenberg via BOCPD; dissipation-as- feature); Phase 3 adds an **encoding-density / data-re-uploading path** (§5.2) and a **sparse/tensor backend** (§5.5). The engine is pure NumPy; an explicit PennyLane circuit reproduces it to ≈5×10^{-16} and compiles, at n=8, to **380 native gates** (220 two-qubit + 160 one-qubit, 20 Trotter layers) — consistent with the ≈50%-connected graph.

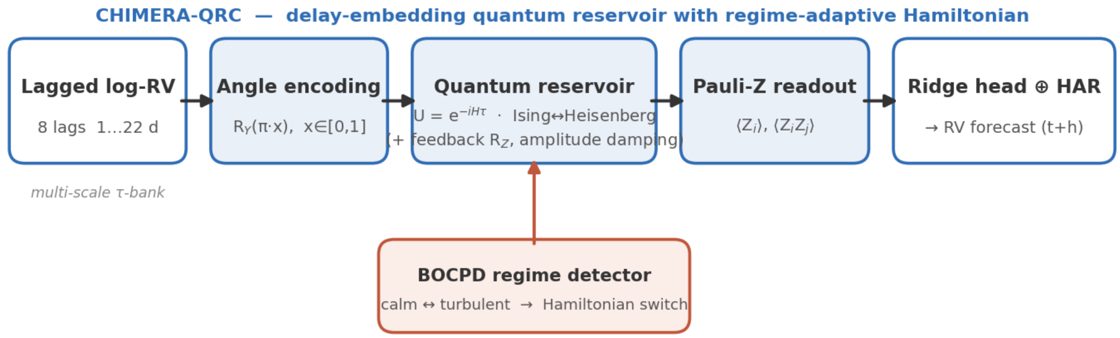

Figure 1. CHIMERA-QRC pipeline: lagged log-RV (and, in Axis B, realized-measure) inputs are angle-encoded onto qubits, evolved under a fixed random-coupled Ising Hamiltonian, and read out as single/pairwise Pauli-Z expectations into a ridge head fused with a linear block of the same inputs; a BOCPD detector selects the Hamiltonian for regime adaptivity.

3. Theoretical and analytical justification

The exploited quantum resources map to RV's structure: (a) **Hilbert-space dimensionality** — n qubits give 2^n amplitudes and $O(n^2)$ measured Pauli features at $O(n^2)$ parameter cost; (b) **intrinsic nonlinearity** from fixed unitary evolution (no gradient training, no barren plateaus); (c) **delay-embedded memory** (the headline model's memory is its explicit lag window, not reservoir recurrence — a recurrent variant is probed in `v3_research`); (d) **Hamiltonian as inductive bias**. The central, falsifiable mechanism is **expressivity scaling**, which we measure directly via the geometric difference (Huang et al. 2021). The quantum kernel is poorly reproducible by a matched classical reservoir: against the name-matched ESN-108 reference, **g(ESN→CHIMERA) ≈ 64 vs ≈3.7 control** (`cli.py run kernel`); against per-n *feature-matched* ESNs the gap is larger but **non-monotonic** (g ≈125 at n=8, peaking ≈158 at n=10, falling ≈88 at n=12). (g is at fixed ridge regularization; the qualitative ~15–40× separation, not the exact value, is the claim.) Crucially, at ≤12–16 qubits the map remains classically simulable, so distinctness is **necessary, not sufficient** for accuracy — and §5.3 shows that, here, it is indeed *not* sufficient.

4. Data modeling strategy

Datasets (all bundled in-repo). Oxford-Man `.SPX` 5-minute realized variance, 2000–2020, 5,029 supervised days incl. the 2008 GFC (Heber et al. 2009; via R packages `highfrequency`/`bvhar`); SPY daily OHLCV 2022–2026 (Garman–Klass proxy; retrieved via Massive.com API, bundled); real MNIST (Keras `.npz` mirror). **Preprocessing:** log-RV target; features min-max scaled to [0,1] on **train only**; per-model ridge penalty selected on a chronological validation tail; multi-seed ensembling. We verified the pipeline is leakage-free (all features lagged ≥1 day; train-only scaling). **Baselines:** HAR, **HAR-X**, GARCH(1,1), GJR-GARCH, AR(3), persistence, ESN (matched + 4×, recurrent), RFF kernel, LSTM. **Metrics:** RMSE(log-RV), QLIKE (Patton 2011), Mincer–Zarnowitz R^2; significance by **Diebold–Mariano with a Newey–West HAC variance** (HLN-corrected), **Holm**-adjusted across the comparison family, and the Model Confidence Set (Hansen–Lunde–Nason 2011).

5. Phase-3 execution — concrete results

Pre-registration (honesty). Before running, we fixed the confirm/refute thresholds in `preregistration.py` (H0/H1/H4, from our Phase-2 §7). All outcomes — including negatives — are reported against them. Full numbers/wall-clock: `results/*_findings.md`.

5.1 The input-bottleneck mechanism (pre-registered negative). With a *fixed* univariate 8-lag encoder, extra qubits receive no new input: g(n) and effective feature-rank **saturate/decline** (n=8→12: g 133→30 at N_sub=800; D_eff 1.8→1.5). **H0 is refuted in this regime** — the empirical case *for* enriching the encoding.

5.2 Encoding density (Axis B) — mechanism confirmed. Feeding the extra qubits genuinely new realized-measure information (signed return/leverage, downside-semivariance share, jump share) restores the lost structure: at n=10, informed vs idle qubits, kernel distinctness and rank jump (**g 52→158, D_eff 1.5→3.1**), and effective rank grows with n (D_eff 1.81→3.14, n=8→12). So the bottleneck is *fixable* — added qubits can carry new information. The decisive question is whether this converts into forecasting **accuracy**.

5.3 The decisive, adversarially-controlled test (the honest headline). We compare CHIMERA to the strongest fair baselines — **HAR-X** (the same rich features used *linearly*, no reservoir), a **true recurrent ESN**, and an **RFF kernel** — all sharing the identical information set and *nesting* HAR-X, so a quantum win requires nonlinearity beyond the linear span. With **8 seeds, HAC-DM, two windows, and Holm correction** (`cli.py run axisB_rig`):

RMSE(log-RV)	HAR-X	recurrent ESN	RFF	CHIMERA	best
crisis n=8	**0.6255**	0.6386	0.6302	0.6379	HAR-X
crisis n=10	0.6034	**0.5996**	0.6047	0.6074	ESN
crisis n=12	**0.5906**	0.6023	0.5964	0.6031	HAR-X
calm n=10	**0.6244**	0.6445	0.6264	0.6291	HAR-X

(plain HAR for reference: crisis 0.6290, calm 0.6454 — HAR-X's rich features cut RMSE sharply.) **HAR-X is best or co-best everywhere; CHIMERA never beats it** and is indistinguishable from the classical ESN/RFF after Holm (one raw loss to RFF at crisis n=8, p=0.049). **After Holm correction no comparison is significant in either direction, and the 95% MCS retains all four models (n=10)** — a statistical tie. The earlier "beats HAR" result was an artifact of a *feature-poor* HAR: the gain is the encoded realized measures (a known SHAR/HARQ effect), not quantum nonlinearity. **By our pre-registered criteria, H0 is refuted — we report this negative.** What honestly survives for the quantum reservoir: it is **competitive** (within ≈0.8–2.1% RMSE of the best at every n), with **lower per-seed dispersion than the recurrent ESN in 3 of 4 cells** (n=12: 0.009 vs 0.015) and **beats it on the calm window** (raw p=0.018; n.s. after Holm). A **tuned, size-unconstrained ESN** (45-config search to 800 nodes) confirms the control was not crippled (0.6020 vs CHIMERA 0.6094, within noise of HAR-X); and **named canonical models** (SHAR, HAR-CJ, HARQ, HEAVY-RM; MSE + QLIKE DM) — **none beats HAR-X**, with CHIMERA tying best on RMSE and only a *raw, non-Holm* QLIKE/MZ edge (`cli.py run canonical`, `esn_tuning_robustness_findings.md`).

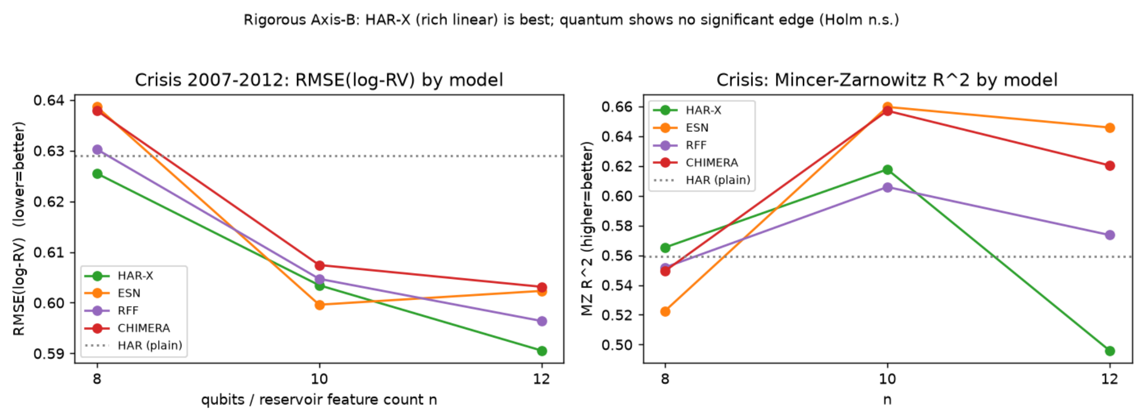

Figure 2. Rigorous Axis-B (8 seeds, crisis window). HAR-X (rich features, linear) is the best or co-best model at every n; the quantum reservoir is competitive but shows no advantage that survives HAC-DM + Holm correction. Left: RMSE(log-RV). Right: Mincer–Zarnowitz R^2.

5.4 Common MNIST benchmark + noise. Same engine, pixels→PCA(n)→n qubits→Pauli readout→ridge. Accuracy **scales with qubits** (n=5/8/10/12: 0.632/0.798/0.831/0.859, 3 seeds; **n=15 sparse-exact continues the trend**, 0.878 vs 0.852 for a paired n=12 on a matched subset — closing the brief's 5/10/15 example) and **beats the linear-PCA baseline at every n** (real nonlinear lift), confirming **sufficient expressivity**; a matched ESN **ties or slightly exceeds** CHIMERA (within ≈1% for n≥8; 2.9% at n=5) — competitive, not dominant. **Noise:** the classifier is **invariant to depolarizing** noise (a uniform Bloch contraction that per-feature standardization removes exactly; accuracy identical across rates 0.05–0.30) and **robust to amplitude damping** (<0.5% at 30%). **Honesty check (**`cli.py run noise_circuit`**):** a per-Trotter-layer density-matrix study shows the *converse* — noise interleaved with the evolution (per-layer single-qubit channels) is **not** removed by standardization (standardized error grows with rate, vs **≈0** for readout-only depolarizing): the invariance is a *readout* property, not circuit-level robustness; accumulated gate error is the real NISQ cost (§6).

5.5 Scaling frontier + quantum-complexity metric. A sparse-exact backend (`expm_multiply`, no dense propagator; matches the dense engine to **2.4×10^{-14}**) reaches **n=16 exactly**. For the random ≈50%-connected reservoir we measure the **bond dimension** χ_eff across a balanced cut: it is **essentially full at every n, χ_eff ≈ 2^(n/2)** (16→255.9 for n=8→16) — an exact MPS gets **zero compression** (S ≈ 1.7–3.2 nats): the reservoir admits **no low-bond-dimension (MPS/TEBD) shortcut**, exact cost stays exponential — the precondition any beyond-frontier advantage would need, though none appears at the scale we can test. **Capability + efficiency checks.** An information-processing-capacity probe (Dambre et al. 2012; `cli.py run capacity`) finds the quantum reservoir **not more** — slightly *less* — nonlinearly expressive than a matched RFF/ESN: no excess expressivity to exploit. A frontier check (`cli.py run frontier`) adds that g(n) does **not** widen toward n=16 (it declines; D_eff/rank grow but a matched ESN keeps pace). **Robustness (supporting study).** *Second family:* a **Heisenberg (XXZ)** reservoir — equally kernel-distinct (g≈55) — also fails to beat HAR-X (0.650 vs 0.642, 8 seeds, full-sample config), so the negative is **not Ising-specific** (`cli.py run second_family`). *Efficiency:* with input held fixed, a *smaller* QRC cannot substitute for a *larger* classical reservoir — quantum accuracy **saturates** while the classical curves improve; per feature the static maps are comparable, so the negative is **saturation, not per-feature inferiority** (weather RMSE °C: CHIMERA 0.85 at its qubit-range ceiling vs RFF 0.78 / recurrent ESN 0.71 at larger budgets). *Domain/architecture:* the *same* engine/protocol on chaotic **weather** (5 stations) and the autonomous **VPT** metric still show no advantage; the **recurrent** QRC is competitive-not-better — unitary evolution is non-dissipative, lacking the contraction behind ESN "generalized synchronization" (Ahmed–Tennie–Magri 2025). We then **fixed the mechanism**: engineered memory-qubit damping traces the pre-registered inverted-U, lifting autonomous VPT ≈+60% to **parity** with the matched ESN — not better (`results/dissipative_qrc_findings.md`).

6. Quantum platform and resource planning

Simulator-first on qBraid: statevector ≤12 qubits, sparse/TN to ≈16, GPU beyond. **Resource estimates** (gate-Trotter, 20 layers): n=8/10/12 use **220/480/760 two-qubit + 160/200/240 one-qubit** gates, yielding **36/55/78 observables** — all Z-basis-diagonal, so **one S-shot set (S≈2–8k, ε≈1/√S) reads every observable** (statevector build ≤~1 min over this range).

QPU validation uses this gate-Trotter circuit on **IonQ / IQM / Rigetti** via qBraid — the random-sparse Ising needs *fewer* two-qubit gates than an all-to-all reservoir, easing NISQ mapping — with **ZNE + measurement mitigation** and a classical cross-check per run. The submission path is **executable now** (`cli.py run qsubmit`): on a simulator it (i) reproduces the engine to **3.9×10^{-16}** (the per-run classical cross-check), (ii) characterizes the **shot budget** (ε≈1/√S; feature error 0.046→0.0066 at S=256→16k; gate-Trotter(20) adds ≈0.04), and (iii) demonstrates **zero-noise extrapolation** under a depolarizing sweep. **Executed on real hardware:** the full mitigated protocol ran first on **Rigetti Cepheus-1-108q** (12 logged cloud jobs; bit order auto-detected; readout 2.3%/6.4%): after lattice routing, scale-1 already exceeds the coherence budget — the accumulated two-qubit-gate cost §5.4 predicts, measured on metal (ZNE marginal; a 4k-shot replication showed the beyond-limit regime stable under day-scale drift). The pre-registered **Garnet protocol** replicated the regime on a second vendor (raw 0.230 > our routing-free 0.171 forecast — **prediction (iii) confirmed**). Cross-platform execution also surfaced a **platform-level finding**: negative-angle rotations are **silently lost server-side on the IonQ route** (RY(π)RY(−π) executed as net RY(π); the client JSON provably contained both), corrupting ZNE's negated folds; we hardened the emitter (mod-2π angles) and validated on-device ($P(|0^8\rangle)=0.996$ vs 0.0). The decisive routing-free **IonQ Forte-1 campaign then executed** under the pre-committed manifest: a smoke gate measured a **2,000-gate/circuit device ceiling** on the native route (corroborated by an OpenQuantum-route probe), so per abort rule it ran scale-1-only (500 shots — a disclosed, cost-forced deviation from the 4k config, s.e.≈0.004; readout 0.08%/0.65%). **Raw 0.104 — below the 0.196 depolarized limit** and every superconducting n=8 number: the **first signal-bearing hardware execution** of this reservoir (**prediction (ii) confirmed**, under our 0.149 forecast; `results/qpu_hardware_findings.md`). A pre-registered **hardware scaling program** (`results/qpu_scaling_outlook.md`) then refuted our own statements on metal (Fig. 3): Garnet turns **signal-bearing at n=10/n=12** while the **same-session seed-0 n=8 anchor** stays scrambled — but a pre-registered cross-seed control (**S7**) found an **independent seed-1 n=8 instance signal-bearing** in the same session, so the n=8 scrambling is **specific to the seed-0 instance, not a size law** (seed-1 n=10 also signal-bearing) — and the newer-generation **Emerald is signal-bearing at the very size Garnet scrambles**, reproduced in a **same-window two-chip pair**: generation-dependent at fixed size, temporally controlled. We thus characterize all four challenge axes — reservoir size, encoding density, shot budget, noise — with the program in one view (raw mean feature error vs size-matched fully-depolarized limit; 4k shots, IonQ 500):

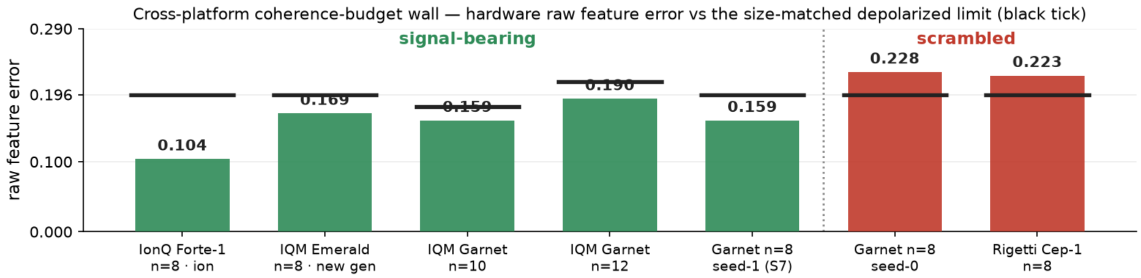

Figure 3. Cross-platform coherence-budget wall: mean raw feature error vs the size-matched depolarized limit (black ticks; 4k shots, IonQ 500). Five configs are signal-bearing (below the limit), two scrambled (Garnet n=8 seed-0 0.228; Rigetti 0.223). The adjacent Garnet n=8 seed-1 (0.159) vs seed-0 (0.228) pair — same chip, same session — refutes a *size* law: the n=8 scrambling is specific to the seed-0 instance. Per-n limits 0.196/0.179/0.214 at n=8/10/12.

Execution provenance (externally verifiable). Every campaign ran against pre-committed predictions — the ten funded ones under a committed manifest (abort/decision rules fixed *before* each launch); every job embeds the repo commit hash and campaign tag in qBraid's timestamped records — a hash-preimage commitment that predictions predate data — and billing matched estimates to the half-credit (`results/CREDIT_BUDGET.md`: ≈64k credits). Bootstrap from the committed counts re-derives every number above and puts **each regime claim 9.7–23.9σ beyond shot noise**; a readout-corrected fingerprint attributes the scrambled regime to predominantly coherent error (`qpu_bootstrap`, `qpu_fingerprint`). Controls were bought where needed: a Rigetti replication (day-drift), a same-session n=8 anchor (drift), a same-window two-chip pair (generation). S5–S6 pre-committed; S7 executed (above).

7. Limitations (stated plainly)

(i) **No quantum advantage** is demonstrated at the ≤16-qubit simulable scale; HAR-X (classical, linear) is the best model on this task. (ii) The RV sample ends Feb-2020 (2008 in-sample, COVID just outside); broader assets/periods are daily-proxy supporting studies (v2_research — same negative), not 5-min RV. (iii) Task-level noise: per-layer single-qubit channels (density-matrix, §5.4) and simulator shot noise; combined noisy-plus-shot execution is measured in the QPU campaigns (§6). (iv) g is regularization- and run-configuration-dependent (qualitative gap only). (v) Superconducting n=8 execution is a **characterized negative** (Fig. 3; day-scale drift exceeds shot noise, so *regime* — not point value — is the robust claim), while n=10/n=12 and the newer generation land inside their limits: the wall is instance- and generation-dependent, not absolute (n=8 scrambling is seed-0-instance-specific — S7). Mitigation recovery on hardware is at best marginal (ZNE 0.223→0.217 on Rigetti; the ion chain is flat — the gate ceiling forbids ZNE). (vi) Distinctness and full-rank entanglement are *necessary, not sufficient* — whether they convert beyond the simulable frontier is open. A quantum-data probe shows the QRC natively reads nonlinear *state* functionals (purity, entanglement) a linear readout cannot — but the proper baseline, **classical shadows** (Huang–Kueng–Preskill 2020), at matched per-state budgets, **wins wherever real information is extracted**, including the shadows-hard $\mathrm{Tr}(\rho^3)$: **closed negatively at simulable scale**; hardware-native many-qubit inputs remain the one untested regime (`results/shadows_hard_findings.md`).

8. Stakeholder impact, milestone plan, AI disclosure

Volatility forecasts feed hedging, risk limits and pricing; we make the impact **concrete** (`cli.py run economics`): a vol-timing backtest on the RV forecast **nearly halves the 2008 max drawdown (−61%→−32%)** — but a **plain HAR** captures it (negative CE fees for the reservoir), so the decision-useful lever is **a simple RV forecast, not quantum hardware**. **Milestone plan:** (i)–(vi) — pre-registration, sweeps, the adversarial test, MNIST+noise, sparse/TN frontier, QPU on three vendors — all ✓ (§6). **AI disclosure:** Claude (Anthropic) assisted with code and drafting; all decisions and results are the team's own.

References

Kornjača et al. 2024 · Zhu et al. 2025 · Ahmed, Tennie & Magri 2025 · Li et al. 2025 · Tandon et al. 2025 · Hou et al. 2025 · Čindrak et al. 2026 · Antoncich et al. 2026 · Kobayashi & Motome 2026 · Huang et al. 2021 · Huang, Kueng & Preskill 2020 (classical shadows) · Dambre et al. 2012 · Corsi 2009 · Patton 2011 · Hansen et al. 2011 · Harvey et al. 1997 · Diebold & Mariano 1995 · Bollerslev 1986 · Jaeger 2001 · Heber et al. 2009.